

Picard-Lefschetz theory part II. speaker: Ivan Tulli

Let $f: \mathbb{C}^n \rightarrow \mathbb{C}$ hol. with isolated crit. point.

$a \in \mathbb{C}^m$ with multiplicity m .

Goal: • associate a group to (f, a) called

the monodromy group $P_f \subset \text{Aut}(\tilde{H}_{n-1}(V_a))$

• Want to describe P_f $\tilde{f}^{-1}(t)$

1. find a basis of $\tilde{H}_{n-1}(V_a)$

vanishing cycles. $\{\Delta_i\}_{i=1}^m$

2. P_f can be described once we know $(\Delta_i \circ \Delta_j)$

1. Monodromy group. (f, a) .

Let $f_\epsilon = f + \epsilon g$ be a Morsification of f

relative to a ball $U \in \mathbb{C}^n$

$\Rightarrow f_\epsilon$ has $\left\{ \begin{array}{l} m \text{ non-deg. critical points. } a_1, \dots, a_n \in U \\ f(q_i) \text{ are distinct.} \end{array} \right.$

for $\epsilon > 0$ sufficiently small.

• $f_\epsilon^{-1}(t) \cap \partial U$ transversely.

Notation. T disk in $\mathbb{C}$ $\{\alpha_1, \dots, \alpha_N\} \subset T$.

$$T' = T \setminus \{\alpha_1, \dots, \alpha_N\}$$

$$V_{T'} = f_E^{-1}(T') \cap U. \quad \partial V_T = f_E^{-1}(T') \cap \partial U$$

Claim. have loc trivial fibrations

$$\begin{array}{ccc} V_{T'} & & \partial V_T \\ \downarrow f_E & & \downarrow f_E \\ T' & & T \end{array}$$

Ehresman fibration
submersion (x crit pt)
+ proper

T' hausdorff
 $K \subset T'$ compact $\Rightarrow K$ closed

$$\Rightarrow f_E^{-1}(K) \text{ closed}$$

$$\Rightarrow f_E^{-1}(K) \cap U \text{ closed bounded subset}$$

$$\text{let } \alpha^* \in T', \quad f_E^{-1}(\alpha^*) \cap U = V_*$$

γ : Loop in T' based at α^*

$$\begin{array}{ccc} V_* & \xrightarrow{\quad} & V_{T'} \\ \downarrow i_0 & \nearrow \tilde{\gamma} & \downarrow f_E \\ V_* \times [0,1] & \xrightarrow{\partial \circ \pi_2} & T' \end{array}$$

$$\begin{array}{ccc} V_{T'} & & \\ \downarrow & \text{fibration} & \\ T' & & \end{array}$$

$\Rightarrow$ homotopy lifting $\tilde{\gamma}$

$$\text{defn: } h_\gamma: V_* \rightarrow V_* \text{ by } h_\gamma = \tilde{\gamma}(-, 1).$$

Note can pick P s.t. $h_\sigma|_{\partial V_\sigma} = \text{id}_{\partial V_\sigma}$

Can induce $h_\sigma: \tilde{H}_{n-1}(V_\sigma) \rightarrow \tilde{H}_{n-1}(V_\sigma)$

$$h_\sigma^r: \tilde{H}_{n-1}(V_\sigma, \partial V_\sigma) \rightarrow \tilde{H}_{n-1}(V_\sigma, \partial V_\sigma)$$

$$\text{Var}_\sigma: \tilde{H}_{n-1}(V_\sigma, \partial V_\sigma) \rightarrow \tilde{H}_{n-1}(V_\sigma)$$

$$\text{Var}_\sigma(\delta) = \delta - h_\sigma \delta$$

Rmk: $h_\sigma, h_\sigma^r, \text{Var}_\sigma$ are indepent of P lift.
only depend on homotopy class of σ .

$$h_{\sigma_\#}(\delta) = \delta + \text{Var}_\sigma \circ \iota_\sigma(\delta) \quad \begin{array}{l} \iota_\sigma: \tilde{H}_{n-1}(V_\sigma) \\ \rightarrow \tilde{H}_{n-1}(V_\sigma, \partial V_\sigma) \end{array}$$

$$h_{\sigma_1 \cdot \sigma_2} = h_{\sigma_1} \cdot h_{\sigma_2}$$

$\Rightarrow$ homomorphism $h_\#: \pi_1(T', \alpha_\sigma) \rightarrow \text{Aut}(\tilde{H}_{n-1}(V_\sigma))$

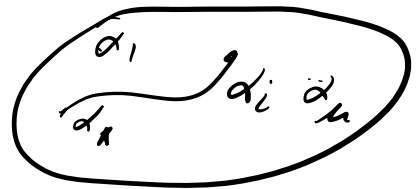
Defn: $\Gamma_f := \text{Im } h_\#$ is called the monodromy group.

Rmk: Independent of morsification.

Vanishing cycles

Let φ a path in T from α_0 to α_i

$$(\varphi[0, 1] \cap \{\alpha_1, \dots, \alpha_N\} = \{\alpha_i\})$$



α_i is nondegenerate critical point of $f|_E$

$$\exists \text{ coords } \underset{\substack{\downarrow \\ \alpha_i}}{\tilde{U}} \subset U \text{ st } f|_E = x_i^2 + z_1^2 + \dots + z_n^2$$

$\Rightarrow$ by last time.

$\tilde{U} \cap f|_E^{-1}(\varphi(t)) \cong$ nbhd of zero section of TS^{n-1}

can find $S_{\varphi(t)}^{n-1} \subset f|_E^{-1}(\varphi(t)) \cap \tilde{U}$ for t close to 1

st $S_{\varphi(t)}^{n-1} \rightarrow$ point as $t \rightarrow 1$

Claim. $\int_{\varphi(t)}^{n-1}$ can be extended to all $t \in [0,1]$

$$\Rightarrow \text{get } [S_x^{n-1}] = \Delta \varphi \in \tilde{H}_{n-1}(V_x)$$

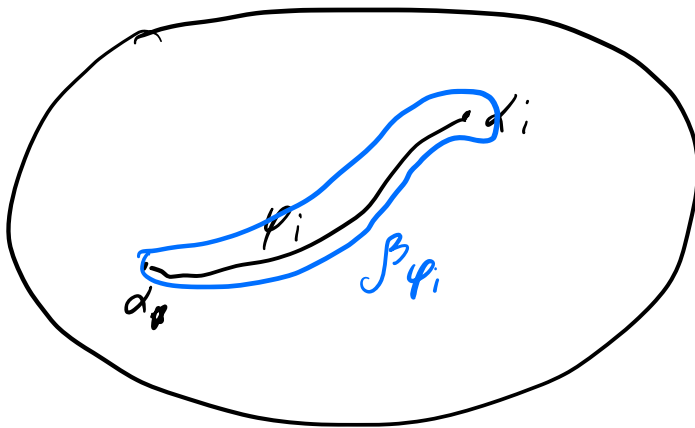
Define. $\Delta \varphi$ is called the vanishing cycles ass. to φ

Rmk. $\Delta \varphi$ is uniquely defed (up to orientatn)
by homotopy class of φ .

Def A set of paths $\{\varphi_i\}_{i=1}^N$ from d_0 to d_i

is (weakly) distinguished. if

the associated loops $\beta \varphi_i$ generate $\pi_1(T', d_*)$



Def: If φ_i is a W.D set paths then
 $\Delta \varphi_i$ is a W.D. basis of vanishing cycles.

Theorem (Briskorn) If $\{\varphi_i\}_{i=1}^N$ A w.D. set of paths
 $\Rightarrow \{\Delta \varphi_i\}$ a basis of $\tilde{H}_{n-1}(V_\alpha)$

Let $\{\varphi_i\}_{i=1}^N$ be a w.D. set of paths.

(Picard-Lefschetz)
Theorem. $h_{\beta_{\varphi_i}^*}(s) = h_{i*}(s) = s + (-1)^{n(n+1)/2} (s \circ \Delta \varphi_i) \Delta \varphi_i$

$P_f = h_*(\pi_1(\underbrace{T'}_{\text{gen. by } \{\beta_{\varphi_i}\}_{i=1}^N}, \alpha_*) \rightarrow \text{gen. by } \{h_{i*}\}_{i=1}^N$

$\rightarrow$ only need to know

$$h_{i*}(\Delta_j)$$

$\Rightarrow$ only need to know

$$\Delta_i \circ \Delta_j$$

$(\Delta_i \circ \Delta_j)$ intersection matrix of (f, q)

Example: ($n=1$)

$w = z^4$ isolated critical point at $z=0$

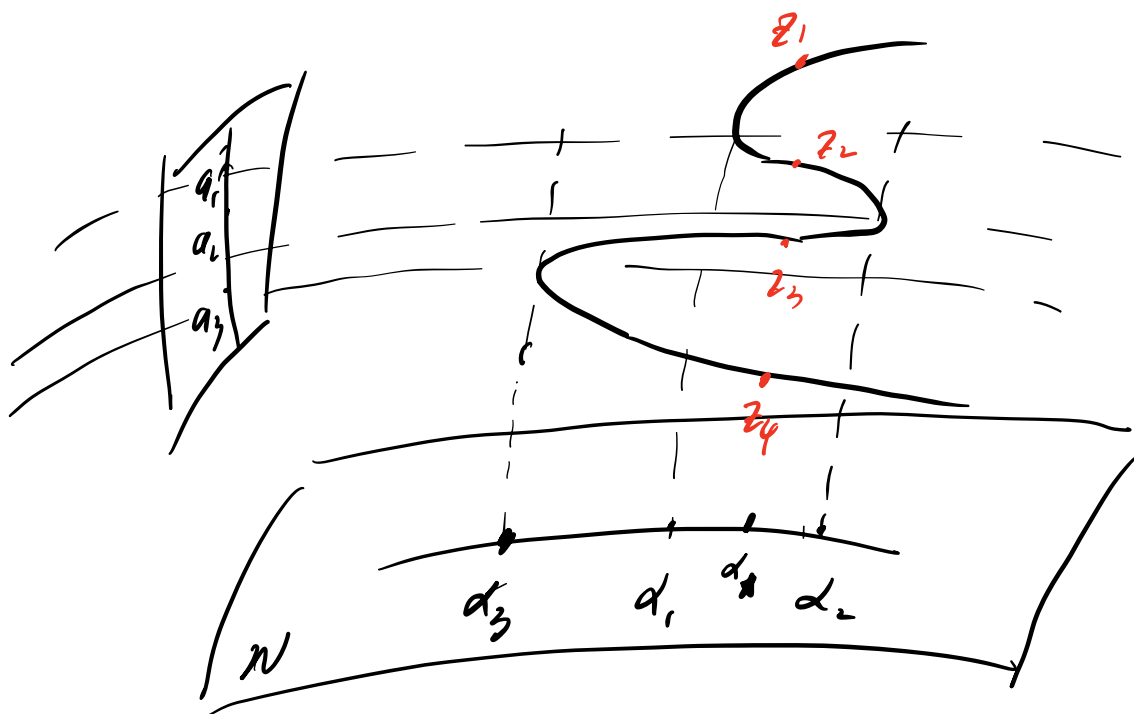
$$N = 3$$

Morsification:

$$w = z^4 - 14z^2 + 24z$$

critical point: $a_1 = 2, a_2 = 1, a_3 = 3$

value $\alpha_1 = 8, \alpha_2 = 11, \alpha_3 = -135$



$$\gamma = \{z_1, z_2, z_3, z_4\}$$

Basis of vanishing cycles $\Delta_i = [z_i] - [z_{i+1}]$

$$(\Delta_i \circ \Delta_j) = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$\begin{aligned} \therefore h_1(\Delta_1) &= \Delta_1 + (-1)^{n(n+1)/2} (\Delta_1 \circ \Delta_1) \Delta_2 \\ &= \Delta_1 - 2\Delta_1 = -\Delta_1 \end{aligned}$$

$$h_1(\Delta_2) = \Delta_2 - (-1) \Delta_1 = \Delta_2 + \Delta_1$$

$$h_1(\Delta_3) = \Delta_3$$

$$\sim h_1 = \begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$